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EGYPT-JAPAN UNIVERSITY OF SCIENCE AND TECHNOLOGY
エジプト日本科学技術大学

Introduction to **Energy**, Environmental and Chemical Engineering

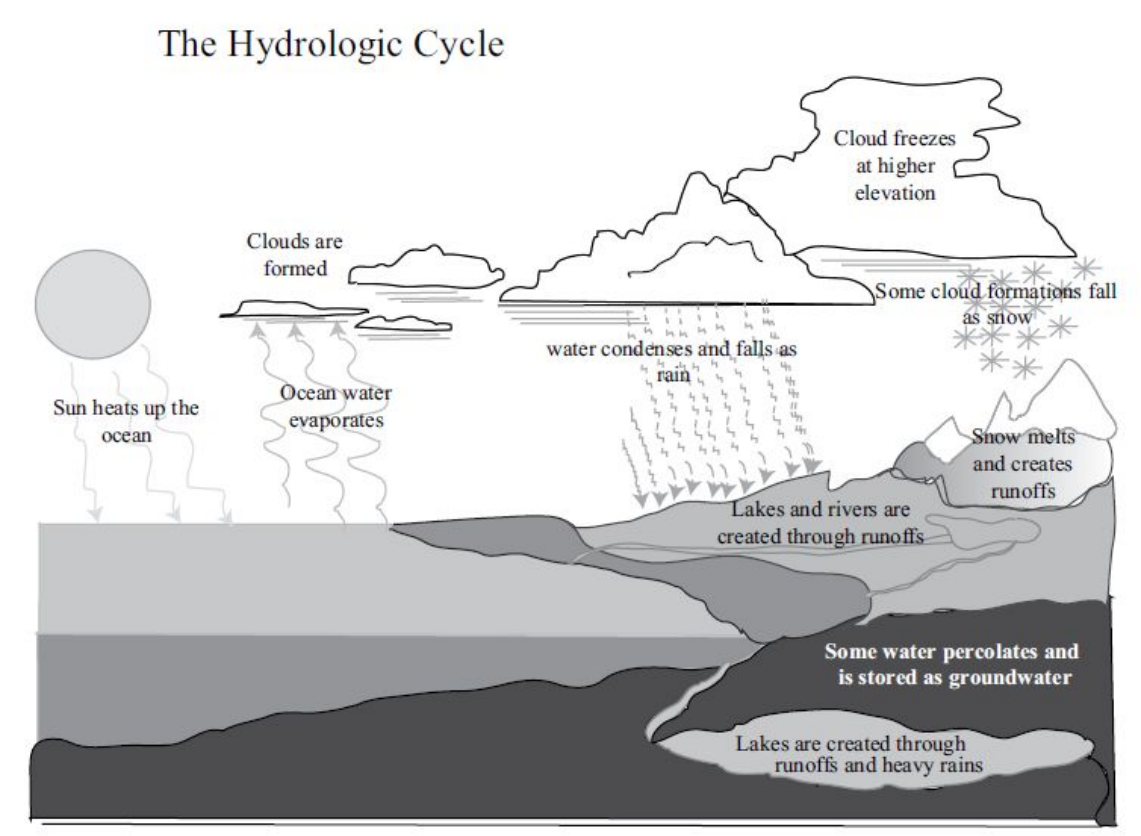
CPE211

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Hydropower

Introduction

- Humankind has harnessed power and energy from water bodies over centuries.
- The hydrologic cycle is one of the earth's fundamental processes—water from the ocean evaporates, turns into clouds, and is brought back to land in the form of rain.



Hydropower

Introduction

- Hydropower supplies about 16% of the world's electricity requirement from all sources.
- Hydropower is the leading renewable energy source for electrical power worldwide and was found to make up 71% of all renewable electricity.
- China is the leading nation in hydropower generation, followed by the United States, Brazil, Canada, India, and then Russia.
- In the United States, there are over 2,000 hydropower plants with a total capacity of over 79 GW.

Hydropower in Egypt



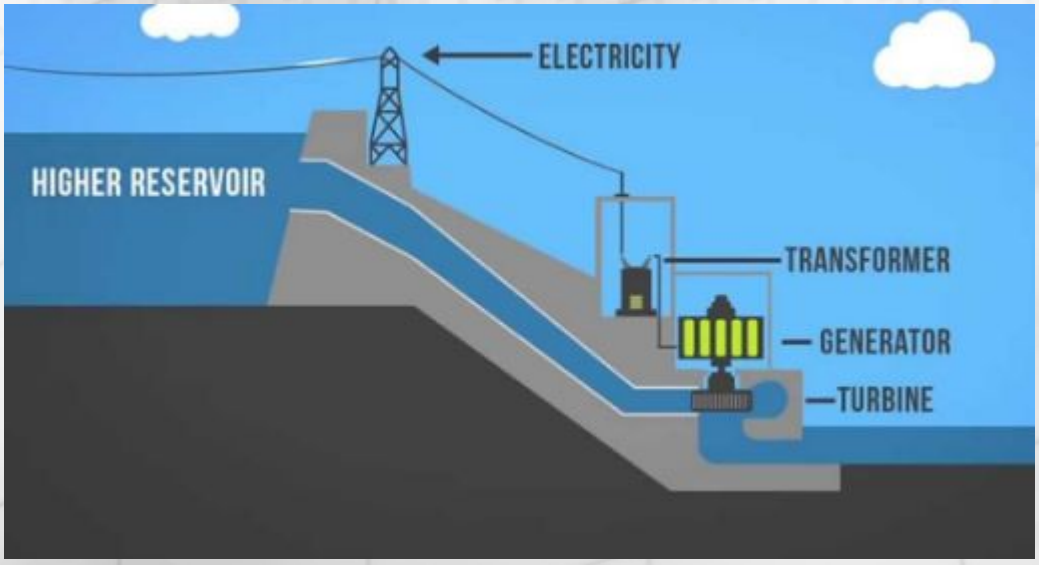
Hydropower plants

Power from Water

The theoretical power derived from water is a consequence of the potential energy of water as it is dropped from a given height.

$$Theoretical\ Power\,(Watts) = Q \times H$$

where
Q = volumetric flow rate (kg/s)
H = dynamic head (meters)



Example:
Determine the theoretical power in watts for water flowing at 200 kg/s and dropped from a dynamic head of 20 meters.

$$P_t\,(Watts) = \frac{200\ kg}{s} \times \frac{20\ m}{} \times \frac{N - s^2}{kg - m} \times \frac{9.8\ m}{s^2} \times \frac{W - s}{N - m} = 39,200\ Watts$$

$$P_t\,(hp) = 39,200\ W \times \frac{hp}{746\ W} = 52.55\ hp$$

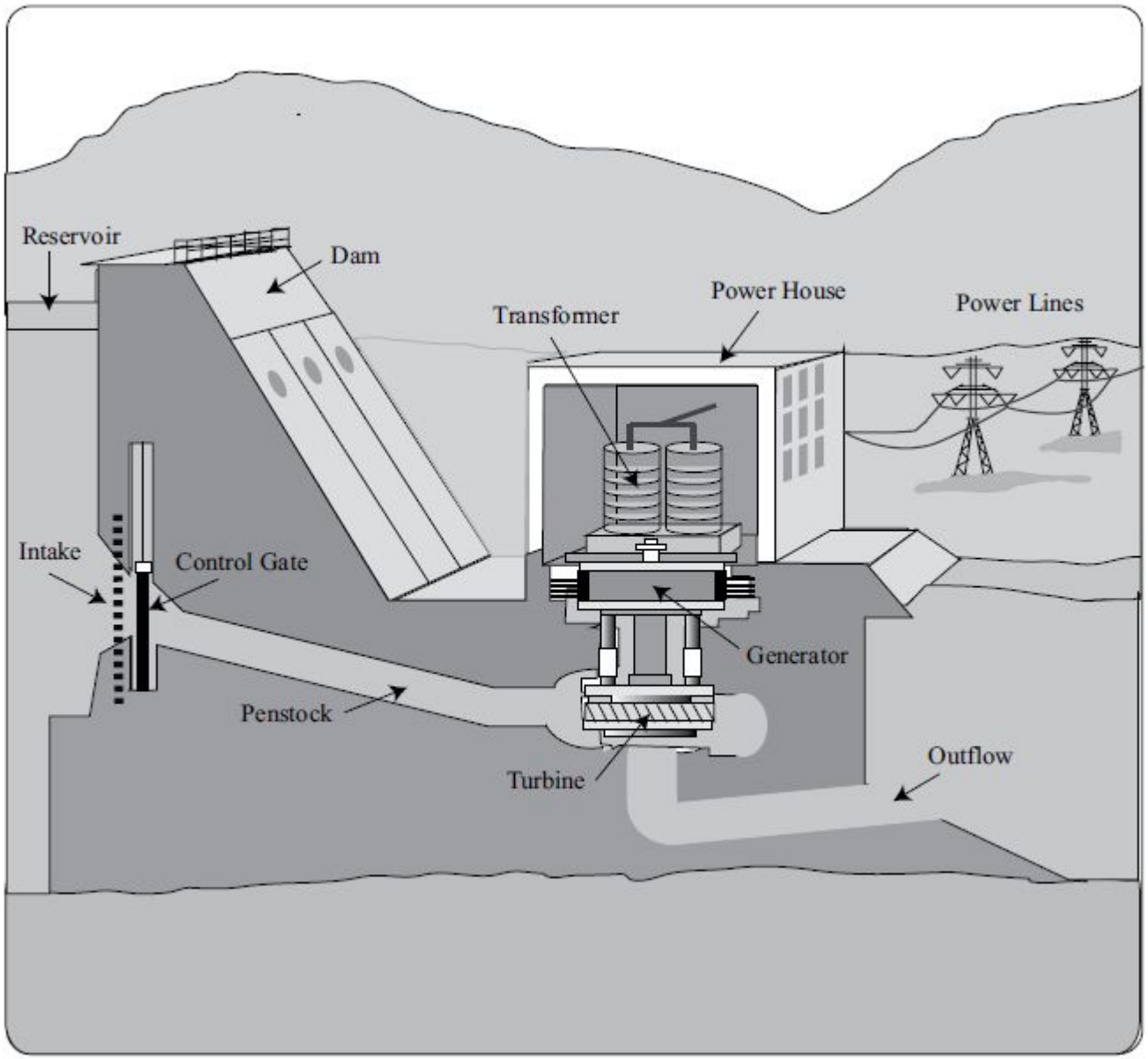
Inefficiencies in Hydro Power Plants

The energy losses in a hydropower plant are mainly due to the following:

a. Hydraulic loses in conduits and turbines
b. Mechanical losses in bearings and the power transmission system
c. Electrical losses in the generator, station use, and transmission (for hydroelectric power)

- The overall effect reduces the theoretical power by a factor of 0.60 to 0.80.
- Large-scale hydro power plants have overall conversion efficiencies of 60% to 80%.
- These values are much higher than those of conventional internal combustion engines and coal power plants.

Basic Components of a Hydro Power Plant



Inefficiencies in Hydro Power Plants

A small commercial micro–hydro power unit had the following data: the volumetric flow rate was 35 liters/s [555 gpm] and the dynamic head was 1.5 meters [4.92 ft]. This unit was reported to produce 200 watts [0.268 hp] of continuous power output at those conditions.
Determine the actual efficiency of this unit.
Also calculate the yearly output of this small device in units of kWh/year, assuming 24 hours per day and 365 days in a year.

The volumetric flow rate is converted into units of kg/s:

$$Q\left(\frac{kg}{s}\right) = \frac{35\ L}{sec} \times \frac{1\ kg}{L} = 35\frac{kg}{s}$$

Then, the theoretical power is calculated:

$$Power(W)_t = Q \times H = \frac{35\ kg}{sec} \times \frac{1.5\ m}{} \times \frac{N - s^2}{kg - m} \times \frac{9.8\ m}{s^2} \times \frac{W - s}{N - m} = 514.5$$

The efficiency is calculated using the ratio of actual and theoretical power as shown:

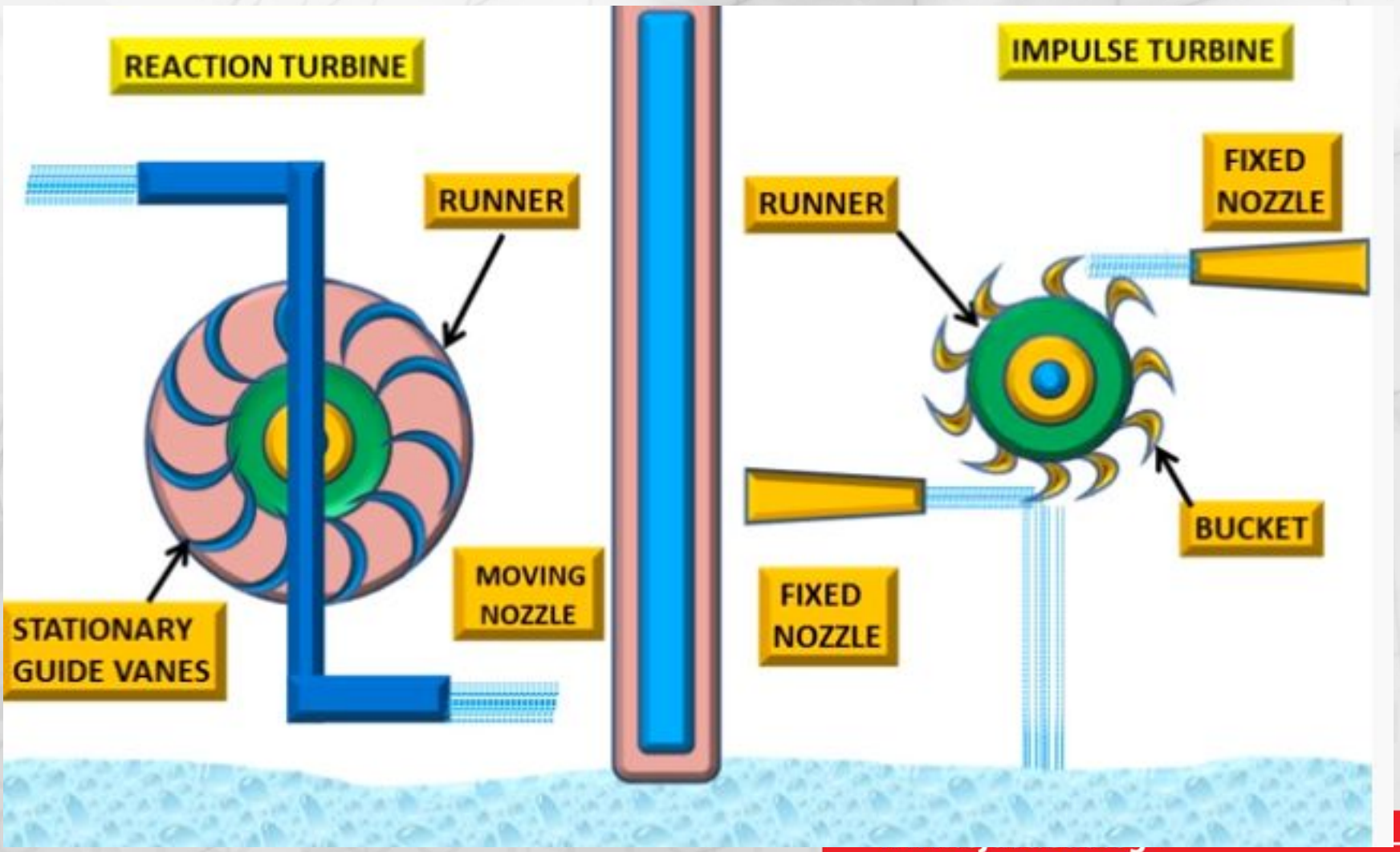
$$\eta(\%) = \frac{Actual\ Power}{Theoretical\ Power} \times 100\% = \frac{200\ W}{514.5\ W} \times 100\% = 38.9\%$$

The yearly output may then be estimated as follows:

$$Yearly\ Output\left(\frac{kWh}{yr}\right) = 200\ W \times \frac{24\ hrs}{day} \times \frac{365\ days}{year} \times \frac{kW}{1{,}000\ W} = 1{,}752\frac{kWh}{yr}$$

Types of Turbines

There are two types of turbines:
a. Impulse turbines b. Reaction turbines



Turbines for Power Generation

The most common types of water turbines for power generation are Francis Turbine, Kaplan Runner, and Pelton Runner



Turbines for Power Generation

a. Francis Turbine. The majority of large hydroelectric power plants use Francis turbines. Typical head ranges for this turbine type are 300 to 500 meters [984 to 1,640 ft]. Specific speeds for these turbines range from 60 to 400 rpm.

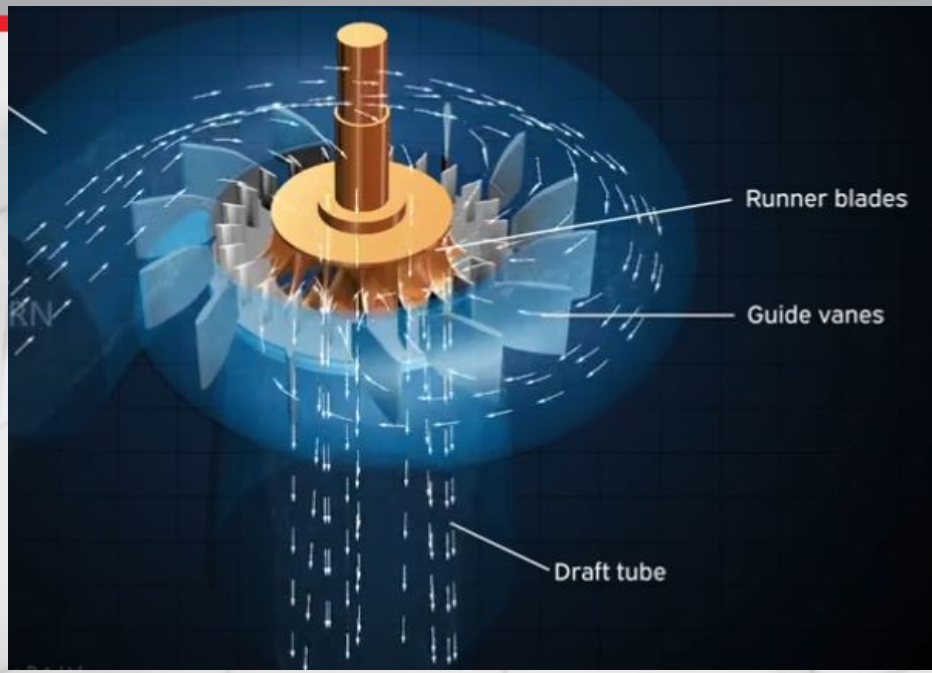
a. Kaplan Runner. The Kaplan runner is a propeller-type turbine, similar to those found in axial flow fans or rotors for pump boats. The dynamic head for Kaplan runners ranges from 2 to 70 meters [6.56 to 229.6 ft]. Its specific speeds range from 300 to 1,100 rpm.

a. Pelton Runner. This is an example of an impulse turbine with blades that look like cups. The water jet is impinged, moving the blades in the opposite direction and turning the shafting where the blades are attached. Pelton runners are very common standby emergency power platforms for hydro power plants. They are usually fed with a pipe of around a fraction of a meter (about 1 ft) and may generate about a MW or a fraction of a MW. The dynamic head for Pelton runners ranges from 50 to 1,300 meters [164 to 4,264 ft]. The specific speeds range from 4 to 70.

Reaction turbines

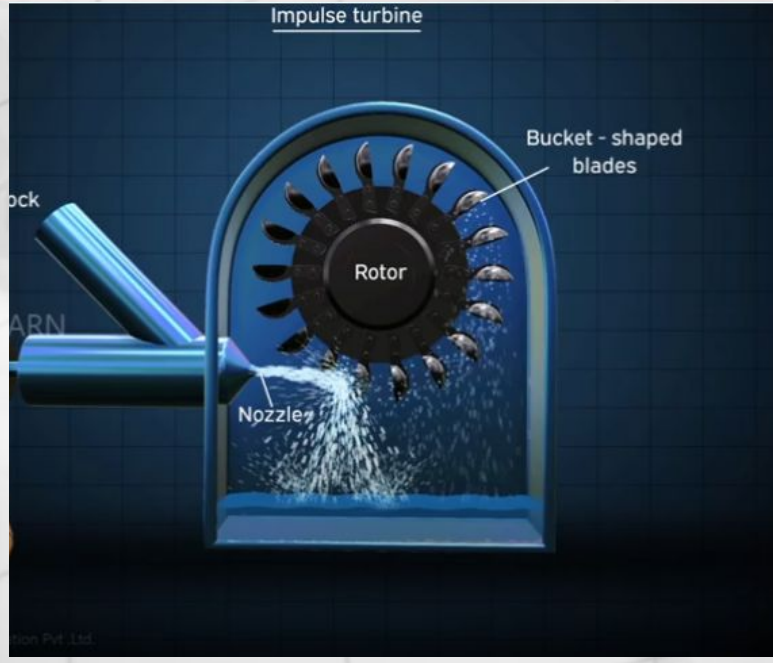
- Reaction turbines, on the other hand, are acted on by water, which changes pressure and gives up energy as it moves through the turbine.

- The turbines must be either encased to contain the water pressure (or suction) or fully submerged in the water flow.



- Newton's Third Law (law of reciprocal reactions or conservation of momentum) describes the transfer of energy for reaction turbines.
- Most reaction turbines are used in low and medium head applications.

Impulse turbines



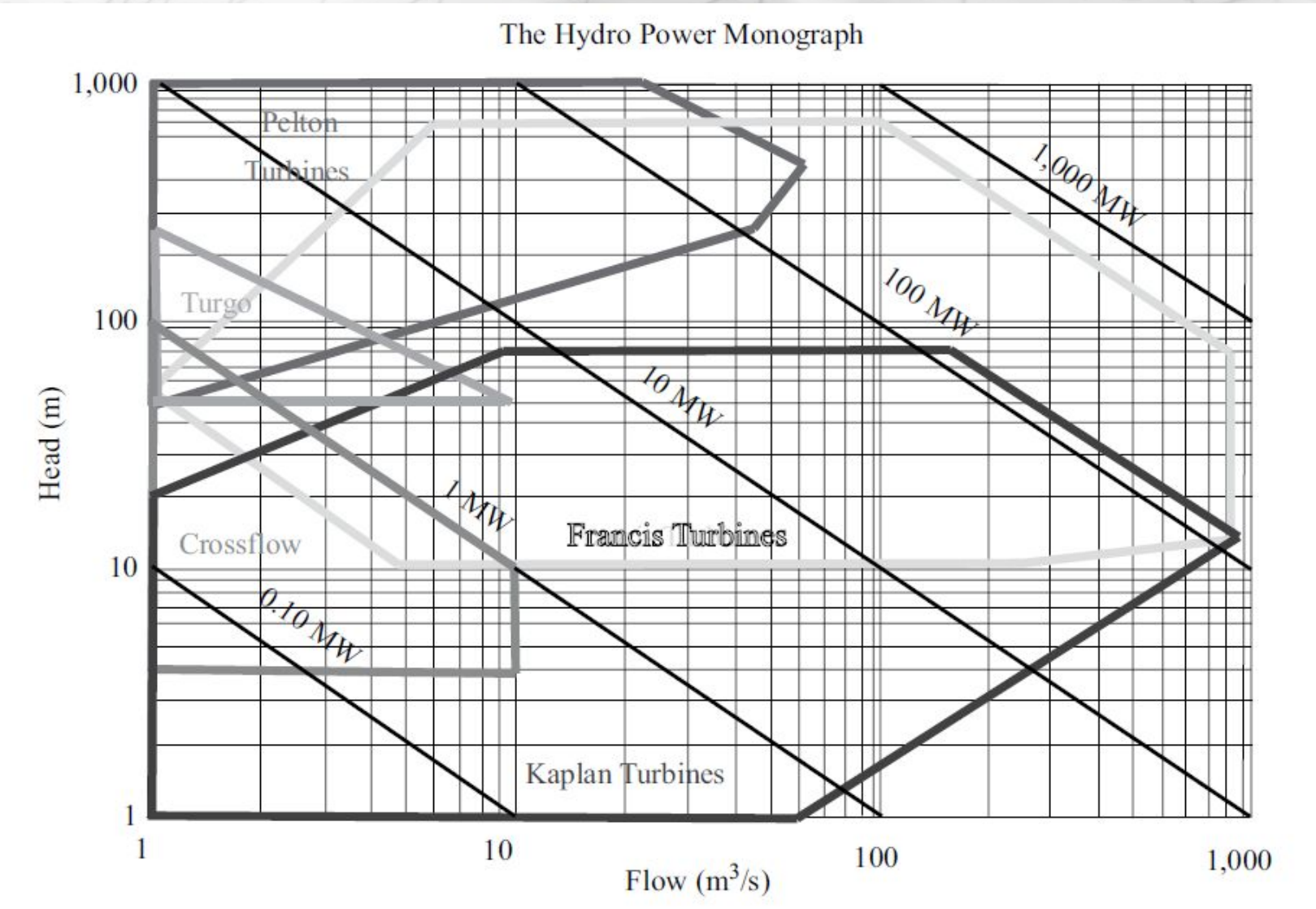
- In impulse turbines, the momentum (or impulse) of water jets is used to move the turbine blades, resulting in a rotary motion of the shafting where the blades are attached.
- Since the turbine is spinning, the force acts over a distance (work), and the diverted water flow is left with diminished energy.

- Newton's Second Law (force = mass × acceleration) describes the transfer of energy in impulse turbines.
- Impulse turbines are most often used in very high head applications.
- Prior to hitting the turbine blades, the water's pressure (its potential energy) is converted into kinetic energy by a nozzle focused on the turbine.
- No pressure change occurs in turbine blades, and the turbine actually does not require housing for this operation.

Turbine Selection

Turbine application chart: a monograph for turbine type selection

Francis and Pelton turbines have the highest potential power output



Specific Speeds for Turbines

- Specific speeds (*Ns*) are used to compare different impellers.

- The specific speed describes the speed of the turbine at its maximum efficiency with respect to the power and flow rate.
- The specific speed is derived to be independent of turbine size.
- If one is given the fluid flow conditions and the desired shaft output speed, the specific speed can be calculated, and an appropriate turbine design selected.

$$N_s = \frac{\omega q^{0.5}}{(gh)^{0.75}}$$

where

ω = pump shaft rotational speed (radians per second)
q = flow rate at the point of best efficiency (m^3/s)
h = total head (m)
g = acceleration due to gravity (9.8 m/s^2)

Specific Speeds for Turbines

Given a pump rotational speed of 1,760 rps, flow rate of 1,500 m_3/s, and head of 100 m, calculate the specific speed.

$$N_s = \frac{\omega q^{0.5}}{(gh)^{0.75}} = \frac{1,760 \times 1,500^{0.5}}{(9.8 \times 10)^{0.75}}$$

$$N_s = \frac{1760 * 1500^{(1/2)}}{(9.8 \times 10)^{(3/4)}} = 2188$$

This is the specific speed of a typical centrifugal pump.

- The specific speeds for various types of pumps and impellers are as follows:
- Radial flow turbines have specific speeds ranging from 500 to 4,000,
- mixed flow turbine speeds range from 2,000 to 8,000,
- and axial flow turbines have specific speeds ranging from 7,000 to 20,000.

Types of Hydro Power Plant

1. On the Basis of Operation

- Based upon its type of operation, a hydro power plant is categorized as either a base load plant or a peak load plant.

- A base load plant operates 24 hours a day, 365 days in a year, and provides continuous power year-round.
- They are designed to operate at this base load indeterminately.
- They are neither allowed to produce power higher than the rated design nor permitted to operate way below their rated capacity.
- Engineers designing these facilities must have several years of running data of their chosen water source, including forecasts for droughts and heavy downpours.

- A hydropower plant may also be designed to operate on a peak load basis.
- It would operate for staggered periods, timed with peak loads during a given season or for a given time of day.
- Some mega-cities use too much power at the beginning of a day and require a dedicated peak load power plant to address their intermittent needs.
- Some peak loads are twice the average requirement and would take quite the toll on a power plant not designed to increase production at such a rate.

Types of Hydro Power Plant

2. Based on Plant Capacity

- Micro-capacity (<5 MW),
- Medium-capacity (5 to 100 MW),
- High-capacity (100 to 1,000), and
- Super-power plant (>1,000 MW)

3. Based on Head

- A low head power plant is said to have a head less than 15 meters [49.2 ft],
- Medium head hydroelectric power plants have dynamic heads between 15 and 50 meters [49.2 to 164 ft],
- High head plants are those with dynamic heads greater than 50 meters.

Types of Hydro Power Plant

4. Based on Hydraulic Features

Conventional

- Conventional hydroelectric power plants generate electrical power using conventional means, taking advantage of the potential and kinetic energy of falling water bodies.
- Water is neither recycled nor reused for subsequent power production and instead is allowed to take its course down the level.
- Dams are usually built through rivers and streams.
- The stored water is used to generate power continuously throughout the year.

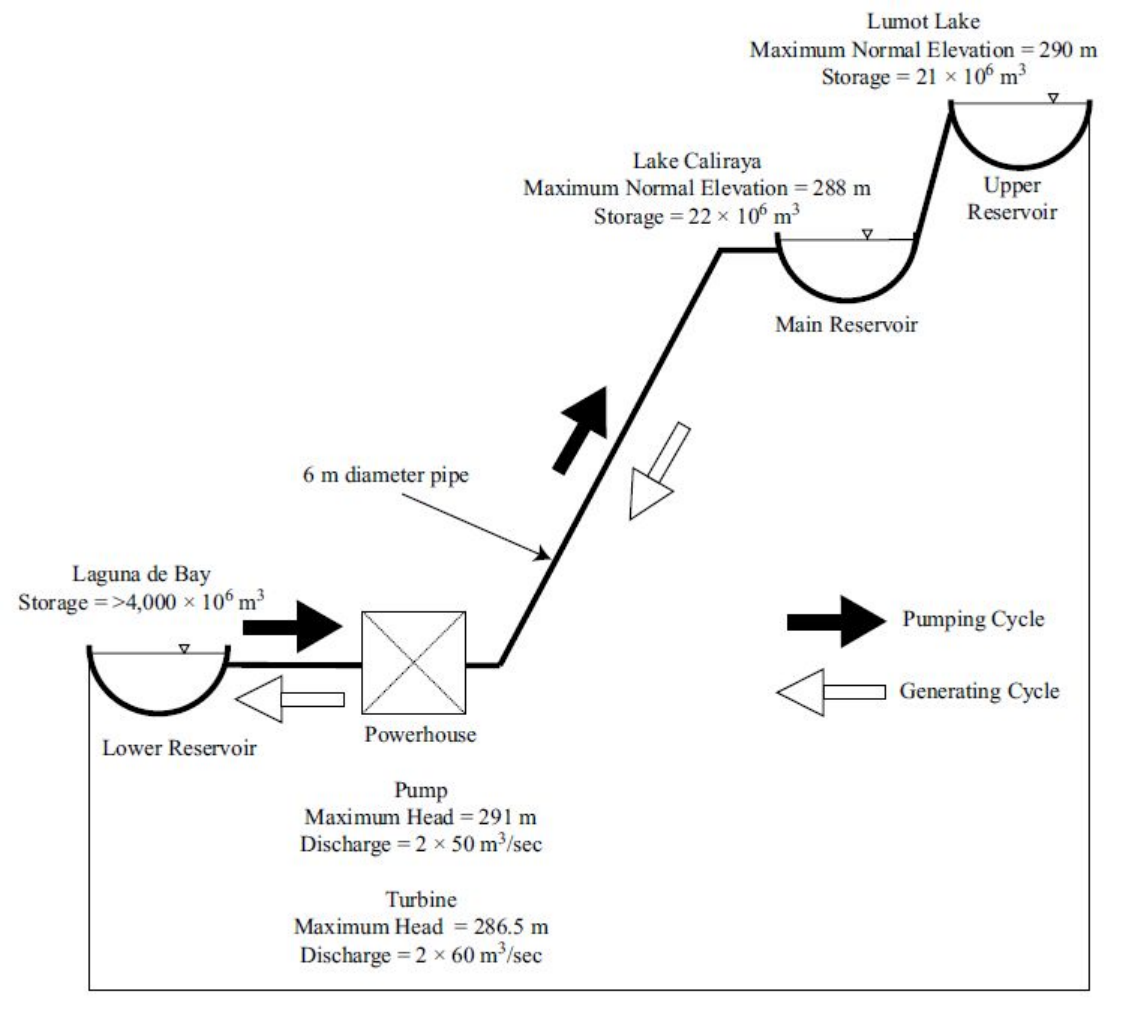
Pumped Storage Systems

- Pumped storage hydro power plants are used in electric power systems to address peak load problems.
- In mega-cities, power demands skyrocket at the beginning of the workday.
- During this time, the demand for power is at its maximum, and if there is no excess power available, brownouts and outages can occur.
- A pumped storage system is built to ensure that when peak power conditions are experienced, it can deliver the needed power.
- Once the power requirement goes down, the water released to generate power is brought back to storage in preparation for the next peak power demand.

Types of Hydro Power Plant

4. Based on Hydraulic Features

Pumped Storage Systems



Turbine Selection

Determine the total maximum time (in hrs and days) for the water from a lake to be emptied when power is generated continuously at the facility. The volumetric flow rate is 280 cubic meters per second [4,434,375 gpm], and Lake has a storage volume of 22×10^6 m^3 [5,807 million gallons].

$$Time(hrs) = \frac{Storage\ Volume(m^3)}{Volumetric\ Flow\ Rate\left(\frac{m^3}{s}\right)} \times \frac{1\ hr}{3{,}600\ s} \times \frac{22 \times 10^6\ m^2}{280\frac{m^3}{s}} \times \frac{1\ hr}{3{,}600\ s} = 21\ hrs$$

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